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path	length
1-2	10
1-4	30
1-2-3	50
1-5	60

* Dynamic programming:

→ Dynamic programming is an algorithm design method that can be used when there is a solution for problem can be viewed as the result of sequence of decisions.

Ex: Knapsack problem

→ The solution to the knapsack problem can be viewed as the result of sequence of decisions.

→ We have to decide the values of x_i, $1 \le i \le n$. First we make a decision on x_1, then on x_2, then on x_3..... So on. A optimal sequence of decisions maximizes the objective function $\sum P_i x_i$.

→ For some of the problems that may be viewed in this way. an optimal sequence of decisions can be formed by making the decisions one at a time and never making wrong decision. This is true for all problems that can be solved by greedy method

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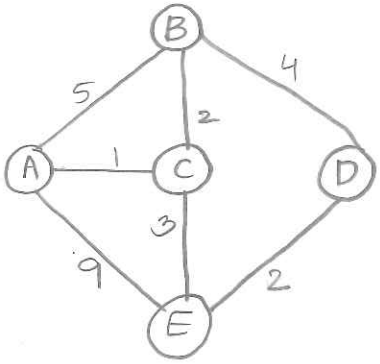
→ For many other problems, it is not possible to make step-wise decisions in such a manner that the sequence of decisions made is optimal.

**

* Principle of optimality:

→ It states that an optimal sequence of decision has the property that whatever the initial state and decision are, the remaining decision must contain an optimal decision sequence with regard to the state resulting from the first decision.

→ For example,



Find the shortest path from vertex A to vertex D

A - C - E - D = 1 + 3 + 2 = 6.

* Comparison between dynamic programming and greedy method:

Dynamic programming	Greedy method.
1. In this method, the solution to a problem can be viewed as result of sequence of	1. Greedy method is the most forward technique for constructing solution to

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decisions	an optimal problem through a sequence of steps.
2. In this method more than one decision is made at a time.	2. Only one decision is made at a time.
3. Dynamic programming considers all possible sequences in order to obtain the optimal solution	3. Greedy method considers an optimal solution without revising the previous solution.
4. It is more expensive than greedy method.	4. Greedy method is comparitively less expensive.

* Comparison between dynamic programming and divide and conquer:

Dynamic programming	Divide and Conquer
→ It is a method in which a solution to a problem can be viewed as the result of sequence of decisions.	→ It is a method in which solution to a problem can be obtained by dividing it into several subproblems.
→ Dynamic programming algorithm solves every sub problem just once and saves the result in a table. From this, solution to the problem is obtained.	→ In divide and conquer, every subproblem is solved and the results are combined to get the original solution..

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→ Dynamic programming works best when all sub problems are dependent.	→ Divide and conquer works best when all sub problems are independent.

* Applications of dynamic programming:

1. Matrix chain multiplication
2. Optimal binary search trees.
3. 0/1 knapsack problem
4. All pairs shortest path problems
5. Travelling sales person problem
6. Reliability design problem.
7. Multi stage graph.

* Multi stage Graph:

→ A multi stage graph $G = (V, E)$ which is a directed graph. In this graph all the vertices are partitioned into K-stages where $K \geq 2$.

→ In multi stage graph problem, we have to find the shortest path from source vertex to destination vertex.

→ The cost of each path is calculated by using the weight given along that edge. The cost of a path from source vertex to destination vertex is the sum of the costs of edges on the path.

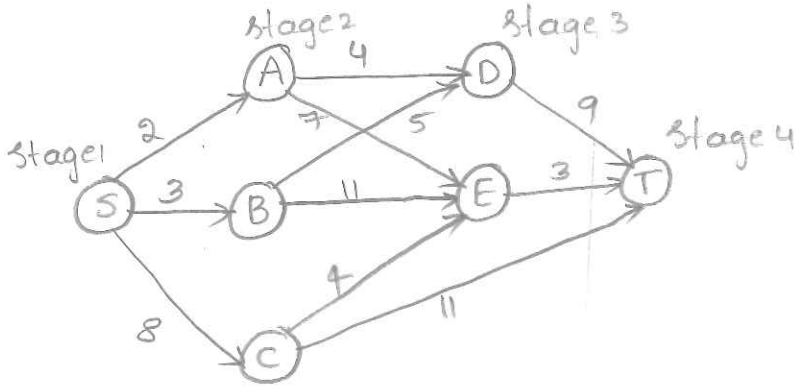
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→ The multi stage graph can be solved using backward approach and forward approach

Backward approach:

→ Consider the following graph



→ There is a single vertex in stage 1, then 3 vertices in stage 2, then two vertices in stage 3 and only one vertex in stage 4 (This is a target stage)

→ c(s,T) where s is source vertex, T is target vertex.

$cost(s,T) = \min\{2 + cost(A,T), 3 + cost(B,T), 8 + cost(C,T)\}$ → eqn ①

→ Now compute cost(A,T), cost(B,T) and cost(C,T)

$cos(A,T) = \min\{4 + cost(D,T), 7 + cost(E,T)\}$ → eqn ②

$cost(B,T) = \min\{5 + cost(D,T), 11 + cost(E,T)\}$ → eqn ③

$cost(C,T) = \min\{4 + cost(E,T), 11\}$

→ Compute cost(D,T) and cost(E,T)

$Cost(D,T) = 9$

$Cost(E,T) = 3$

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→ The Backward vertex is E

→ Substitute these values in eqn ②, ③, ④

$$cost(A,T) = \min\{4+9, 7+3\}$$
$$= \min\{13, 10\}$$
$$= 10$$

$$cost(B,T) = \min\{5+9, 11+3\}$$
$$= 14$$

$$cost(C,T) = \min\{4+3, 11\}$$
$$= 7.$$

∴ Therefore, cost(S,T) is

$$cost(S,T) = \min\{2+cost(A,T), 3+cost(B,T), 8+cost(C,T)\}$$
$$= \min\{2+10, 3+14, 8+7\}$$
$$= \min\{12, 17, 15\}$$

$cost(S,T) = 12$

→ The path with minimum cost is S – A – E – T with the cost 12.

→ This method is known as backward approach.

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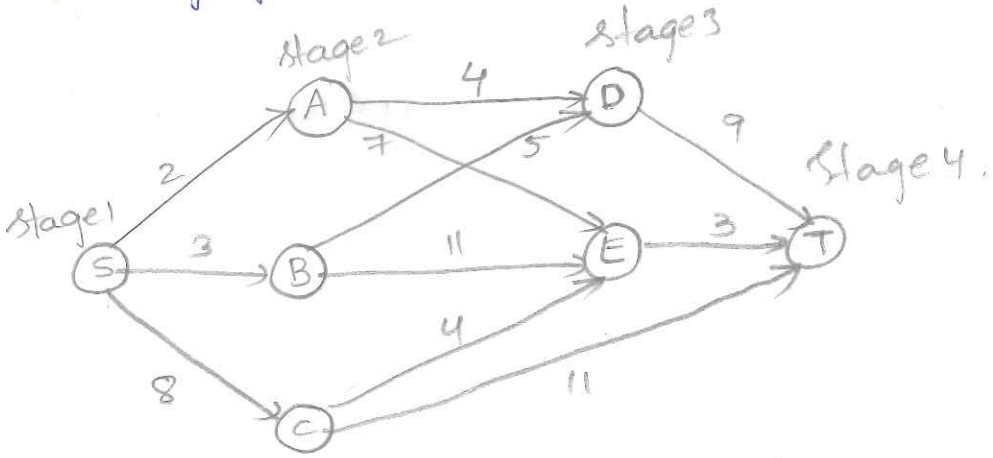
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Algorithm for backward approach:

```
Algorithm BGraph (G, k, n, p)
{
b cost[i] := 0
for j := 2 to n do
{
let r be such that <r, j> is an edge of
  G and b cost[r] + c[r, j] is minimum
  b cost[j] := b cost[r] + c[r, j];
  d[j] := r;
  }
  P[1] := 1 ; P[k] := n;
  for j := k-1 to 2 do P[j] := d[P[j+1]];
  }
```

* Forward approach:

→ Find the shortest path between source vertex and destination vertex using forward approach for given graph.



cost(S, A) = 2

cost(S, B) = 3

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$cost(S,C) = 8$

$Cost(S,D) = \min\{2 + cost(A,D), 3 + cost(B,D)\}$

$= \min\{2+4, 3+5\}$

$= 6.$

$cost(S,E) = \min\{2 + cost(A,E), 3 + cost(B,E), 8 + cost(C,E)\}$

$= \min\{2+7, 3+11, 8+4\}$

$= 9.$

$Cost(S,T) = \min\{cost(S,D) + cost(D,T), cost(S,E) + cost(E,T),$

$cost(S,C) + cost(C,T)\}$

$= \min\{6+9, 9+3, 8+11\}$

$Cost(S,T) = 12$

→ The minimum cost is 12 with path S-A-D-F. This method is called forwaed approach. 2)

→ Time complexity for forwaed approach is $\Theta(|V|+|E|)$

→ Time Complexity for backwaed approach is $\Theta(|V|+|E|)$.

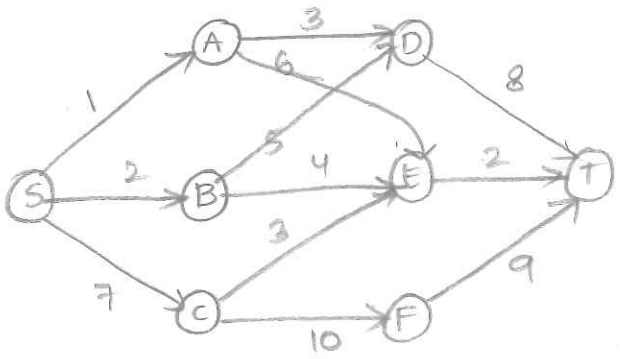
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Algorithm for forward approach:

```
Algorithm fGraph (G, k, n, p)
{
cost[n] := 0.0;
for j = n-1 to 1 step do
{
let r be a vertex such that <j, r> is an edge
of G and C[j, r] + cost[r] is minimum
    cost[j] := c[j, r] + cost[r];
    d[j] := r;
    }
    P[1] := 1; P[k] := n;
    for j := 2 to k-1 do p[j] := d[p(j-1)];
    }
```

his 2)



· Find using backward approach.

$cost(S,T) = \min\{1 + cost(A,T), 2 + cost(B,T), 7 + cost(C,T)\}$

$cost(A,T) = \min\{3 + cost(D,T), 6 + cost(E,T)\}$

$= \min\{3+8, 6+2\}$

$= 8$

$cost(B,T) = \min\{5 + cost(D,T), 4 + cost(E,T)\}$

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$= \min\{5+8, 4+2\}$

$= 6$

$cost(C,T) = \{\min\{3+cost(E,T), 10+cost(E,T)\}$

$= \min\{3+2, 10+9\}$

$= 5$

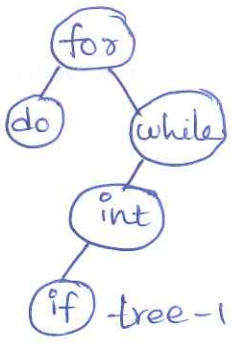
$cost(S,T) = \{1+8, 2+6, 7+5\}$

$= 8$

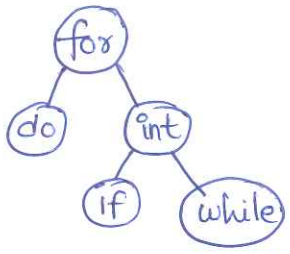
path is S-B-E-T.

*Optimal binary search tree:

→ Optimal binary search tree is a binary search tree for which the nodes are arranged such that the tree cost is minimum. Consider the 2 following possible BST.



tree-1



tree 2

→ The above tree consist of set of identifiers for, do, while, int, if.

→ we can create different BST for same identifier set.

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→ In the 1st tree, we require 4 comparisons to find the identifier in the worst case.

→ In the 2nd tree, we requires '3' comparisons to find an identifier in the worst case.

→ Average no of comparisons required for 1st tree is

$$\frac{1+2+2+3+4}{5} = \frac{12}{5} = 2.4$$

→ Average no of comparisons required for 2nd tree is

$$\frac{1+2+2+3+3}{5} = \frac{11}{5} = 2.2.$$

→ Second BST is considered as OBST where its cost is 2.2.

→ In a general situation, we can expect different identifiers to be searched for different probabilities. In addition, we can expect unsuccessful searches also to be made.

→ To obtain a cost function for binary search trees it is useful to add external nodes.

→ External nodes are denoted with square symbol.

→ Internal nodes are denoted with circular symbol.

→ If a binary search tree represents 'n' identifiers then there will be exactly 'n' internal nodes and

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'n+1' external nodes.

→ Every internal node represents a point where a successful search may terminate.

→ Every external node represents a point where an unsuccessful search may terminate.

→ The following formula is used to find the cost of a binary search tree.

$$\sum_{1 \le i \le n} P(i) * level(a_i) + \sum_{0 \le i \le n} q(i) * (level(E_i) - 1)$$

where P(i) is probability of internal nodes

a_i $(a_1, a_2, \ldots a_n)$ represents internal nodes.

q(i) is probability of external nodes

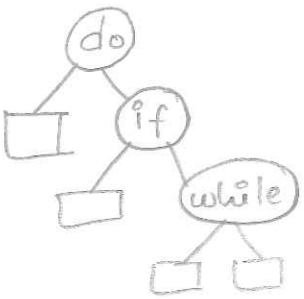
E_i $(E_1, E_2, \ldots E_n)$ represents external nodes

10) Consider the identifier set $(a_1, a_2, a_3) = (do, if, while)$ and probabilities of internal nodes are $P(1) = 0.5$, $P(2) = 0.1$, $P(3) = 0.05$ and probabilities of external nodes are $q(0) = 0.15$, $q(1) = 0.1$, $q(2) = 0.05$, $q(3) = 0.05$. Construct possible binary search trees and find OBST.

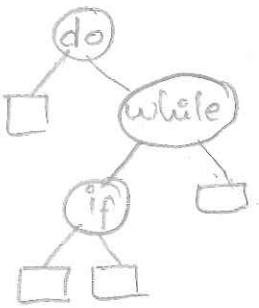
Sol:- Construct possible BST's for the given identifiers (do, if, while).

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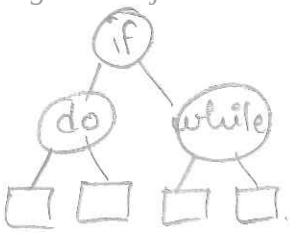
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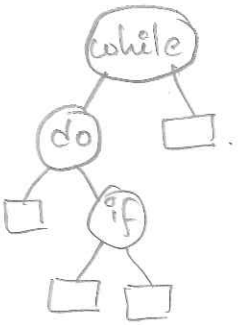
tree 1



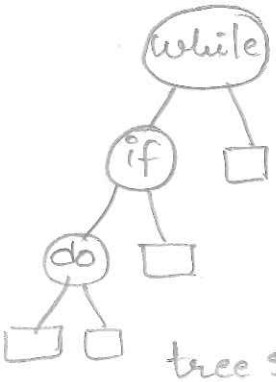
tree 2



tree 3



tree 4



tree 5

cost(tree 1) = $[P(1).\ level(a_1) + P(2).\ level(a_2) + P(3).\ level(a_3)]$

$+[q(0)\ level(\varepsilon_0 - 1) + q(1)\ level(\varepsilon_1 - 1) + q(2)\ level(\varepsilon_2 - 1)$

$+ q(3)\ level(\varepsilon_3 - 1)]$

$= [0.5 \times 1 + 0.1 \times 2 + 0.05 \times 3] + [0.15 \times 1 + 0.1 \times 2 + 0.05 \times 3 + 0.05 \times 3]$

$= 0.85 + 0.65$

$= 1.5.$

cost(tree2) $= (0.5 \times 1 + 0.1 \times 3 + 0.05 \times 2) + (0.15 \times 1 + 0.1 \times 2 + 0.05 \times 3 + 0.05 \times 3)$

$= 0.9 + 0.65$

$= 1.55$

cost(tree3) $= (0.5 \times 2 + 0.1 \times 2 + 0.05 \times 2) + (0.15 \times 2 + 0.1 \times 2 + 0.05 \times 2 + 0.05 \times 2)$

$= 1.9$

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cost (tree 4) = [0.5x2 + 0.1x3 + 0.05x1] + [0.15x1 + 0.1x2 + 0.05x3 + 0.05x3]

= 1.35 + 0.65.

= 2.

cost (trees) = [0.5x3 + 0.1x2 + 0.05x1] + [0.15x1 + 0.1x2 + 0.05x3 + 0.05x3]

= (0.5 + 0.2 + 0.05) + (0.65)

= 1.75 + 0.65

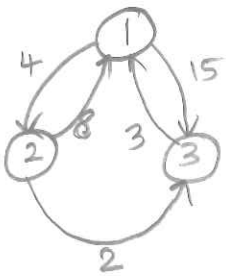
= 2.4

1.5
0.2
0.05
1.75
0.65
2.40

*All pairs shortest path problem:

→ The problem of determining the shortest path among all pairs of vertices of a given graph is

Consider the given graph



→ The above graph consisting of three vertices 1, 2, 3. So we have to find shortest paths b/w all types of vertices 1, 2, 3.

→ All pairs are A(1,1), A(1,2), A(1,3), A(2,1), A(2,2), A(2,3), A(3,1), A(3,2), A(3,3)

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Step-1 → ~~In adjacen~~ Represent the given matrix in adjacency matrix form.

$$A^0 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 4 & 15 \\ 8 & 0 & 2 \\ 3 & \infty & 0 \end{bmatrix} \end{array}$$

$$A^0 = w[i,j]$$

$$A^k_{(i,j)} = \min\left\{ A^{k-1}_{(i,j)},\ A^{k-1}_{(i,k)} + A^{k-1}_{(k,j)} \right\}$$

Step-2: Take vertex 1 as intermediate vertex ($k=1$) and find shortest path among all pairs of vertices

$i=1, j=1, k=1$

$$A^1_{(1,1)} = \min\left\{ A^{k-1}_{(1,1)},\ A^{k-1}_{(1,1)} + A^{k-1}_{(1,1)} \right\}$$

$$= \min\left\{ A^0_{(1,1)},\ A^0_{(1,1)} + A^0_{(1,1)} \right\}$$

$$= \min\{0, 0+0\}$$

$$= 0$$

$i=1, j=2, k=1$

$$A^1_{(1,2)} = \min\left\{ A^0_{(1,2)},\ A^0_{(1,1)} + A^0_{(1,2)} \right\}$$

$$= \min\{4, 0+4\}$$

$$= 4$$

$i=1, j=3, k=1$

$$A^1_{(1,3)} = \min\left\{ A^0_{(1,3)},\ A^0_{(1,1)} + A^0_{(1,3)} \right\}$$

$$= \min\{15, 0+15\}$$

$$= 15$$

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Step-1

→ ~~In adjacen~~ Represent the given matrix in adjacency matrix form.

$$A^0 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 4 & 15 \\ 8 & 0 & 2 \\ 3 & \infty & 0 \end{bmatrix} \end{array}$$

$$A^0 = w[i,j]$$

$$A^k_{(i,j)} = \min\left\{ A^{k-1}_{(i,j)},\ A^{k-1}_{(i,k)} + A^{k-1}_{(k,j)} \right\}$$

Step-2: Take vertex 1 as intermediate vertex ($k=1$) and find shortest path among all pairs of vertices

$i=1, j=1, k=1$

$$A^1_{(1,1)} = \min\left\{ A^{k-1}_{(1,1)},\ A^{k-1}_{(1,1)} + A^{k-1}_{(1,1)} \right\}$$

$$= \min\left\{ A^0_{(1,1)},\ A^0_{(1,1)} + A^0_{(1,1)} \right\}$$

$$= \min\{0, 0+0\}$$

$$= 0.$$

$i=1, j=2, k=1$

$$A^1_{(1,2)} = \min\left\{ A^0_{(1,2)},\ A^0_{(1,1)} + A^0_{(1,2)} \right\}$$

$$= \min\{4, 0+4\}$$

$$= 4$$

$i=1, j=3, k=1$

$$A^1_{(1,3)} = \min\left\{ A^0_{(1,3)},\ A^0_{(1,1)} + A^0_{(1,3)} \right\}$$

$$= \min\{15, 0+15\}$$

$$= 15$$

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$i=2, j=1, k=1$

$A^1_{(2,1)} = \min\{A^0_{(2,1)}, A^0_{(2,1)} + A^0_{(1,1)}\}$

$= \min\{8, 8+0\}$

$= 8$

$A^1_{(2,2)} =$

$i=2, j=2, k=1$

$= \min\{A^0_{(2,2)}, A^0_{(2,1)} + A^0_{(1,2)}\}$

$= \min\{0, 8+4\}$

$= 0.$

$A^1_{(2,3)} =$

$i=2, j=3, k=1$

$= \min\{A^0_{(2,3)}, A^0_{(2,1)} + A^0_{(1,3)}\}$

$= \min\{2, 8+15\}$

$= 2.$

$A^1_{(3,1)} =$

$i=3, j=1, k=1$

$= \min\{A^0_{(3,1)}, A^0_{(3,1)} + A^0_{(1,1)}\}$

$= \min\{3, 3+0\}$

$= 3$

$A^1_{(3,2)} =$

$i=3, j=2, k=1$

$= \min\{A^0_{(3,2)}, A^0_{(3,1)} + A^0_{(1,2)}\}$

$= \min\{\infty, 3+4\}$

$= 7$

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$A^1_{(3,3)} \Rightarrow i=3, j=3, k=1$

$= \min(0, 3+15)$

$= 0.$

$$A^1 = \begin{bmatrix} 0 & 4 & 15 \\ 8 & 0 & 2 \\ 3 & 7 & 0 \end{bmatrix}$$

Step-3: Take vertex '2' as a intermediate and find shortest path between all pairs of vertices.

$A^2_{(1,1)} = \min\{0, 4+8\} = 0$

$A^2_{(1,2)} = \min\{4, 4+0\} = 4$

$A^2_{(1,3)} = \min\{15, 4+2\} = 6.$

$A^2_{(2,1)} = \min\{8, 0+8\} = 8$

$A^2_{(2,2)} = \min\{0, 0+0\} = 0.$

$A^2_{(2,3)} = \min\{2, 0+2\} = 2$

$A^2_{(3,1)} = \min\{3, 7+8\} = 3$

$A^2_{(3,2)} = \min\{7, 7+0\} = 7$

$A^2_{(3,3)} = \min\{0, 7+2\} = 0.$

$$A^2 = \begin{bmatrix} 0 & 4 & 6 \\ 8 & 0 & 2 \\ 3 & 7 & 0 \end{bmatrix}$$

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Step-4: Take vertex '3' as intermediate and find shortest path between all pairs of vertices.

$A^3_{(1,1)} = \min\{0, 6+3\} = 0$

$A^3_{(1,2)} = \min\{4, 6+7\} = 4$

$A^3_{(1,3)} = \min\{6, 6+0\} = 6$

$A^3_{(2,1)} = \min\{8, 2+3\} = 5$

$A^3_{(2,2)} = \min\{0, 2+7\} = 0$

$A^3_{(2,3)} = \min\{2, 2+0\} = 2$

$A^3_{(3,1)} = \min\{3, 0+3\} = 3$

$A^3_{(3,2)} = \min\{7, 0+7\} = 7$

$A^3_{(3,3)} = \min\{0, 0+0\} = 0$

$$A^3 = \begin{bmatrix} 0 & 4 & 6 \\ 5 & 0 & 2 \\ 3 & 7 & 0 \end{bmatrix}$$

* 0/1 knapsack problem:

→ It contains n items with profits $(P_1, P_2, \ldots P_n)$ and weights $(w_1, w_2, \ldots w_n)$. knapsack means empty bag it has a capacity of m units. and the objective

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function of this problem is to get maximum profit by filling empty back with given items. Subject to constraints $0 \le x_i \le 1$ and $m \le \Sigma w_i$, $i \le n$.

→ we can solve this problem by using dynamic programming method

→ It uses the following steps.

Step-1:

Computational formula

$$S_1^i = S^{i-1} + (P_i, w_i)$$

→ It is used to add tuples

$$S^i = S^{i-1} + S_1^i$$

→ This formula is used for merging operation.

Step-2:

purging rule

→ It decides which tuple is to be discard and which tuple is to be accept

→ If one of S^{i-1} and S_1^i has a pair (P_j, w_j) and other (P_k, w_k) and $P_j \le P_k$, $w_j > w_k$ then discard (P_j, w_j)

Step-3:

Selection of items:

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In this step, we can decide which item is to be placed in the bag based on following constraints

if $(P_i, w_i) \in s^n$ & $(P_i, w_i) \notin s^{n-1}$ then $x_n = 1$ else $x_n = 0$.

Problems:

1) Consider the instance $n=3$, $m=6$, $(P_1, P_2, P_3) = (1,2,5)$ $(w_1, w_2, w_3) = (2,3,4)$. Find the maximum profit by using dynamic programming method.

Sol:- Step-1: Initially

$$S_i^0 = (0,0), \quad S^0 = (0,0)$$
$$= (P_i, w_i)$$

$i=1 \Rightarrow S_1^i = S^{i-1} + (P_i, w_i)$

$$= S^0 + (P_1, w_1)$$
$$= (0,0) + (1,2)$$
$$= (1,2)$$

merging $i=1$

$$S^i = S^{i-1} + S_1^i$$
$$S^1 = S^0 + S_1^1$$
$$= (0,0) + (1,2)$$
$$S^1 = \{(0,0), (1,2)\}$$

$i=2 \Rightarrow S_2^i = S^{i-1} + (P_i, w_i)$

$$= S^1 + (P_2, w_2)$$

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$= \{(0,0),(1,2)\} + \{(2,3)\}$

$= \{(2,3),(3,5)\}$

$S^i = S^{i-1} + S_1^i$

$S^2 = S^1 + S_1^2$

$= \{(0,0),(1,2)\} + \{(2,3),(3,5)\}$

$S^2 = \{(0,0),(1,2),(2,3),(3,5)\}$

i = 3

$S_1^i = S^{i-1} + (P_i, W_i)$

$S_1^3 = S^2 + (P_3, W_3)$

$\{(0,0),(1,2),(2,3),(3,5)\} + \{(5,4)\}$

$= \{(5,4),(6,6),(7,7),(8,9)\}$

$S^i = S^{i-1} + S_1^i$

$S^3 = S^2 + S_1^3$

$= \{(0,0),(1,2),(2,3),(3,5)\} + \{(5,4),(6,6),(7,7),(8,9)\}$

$= \{(0,0),(1,2),(2,3),(3,5),(5,4),(6,6),(7,7),(8,9)\}$

Step-2: Apply the purging rule on tuples of

$S^3 = \{(0,0),(1,2),(2,3),(3,5),(5,4),(6,6),(7,7),(8,9)\}$

discard tuples (7,7), (8,9) because tuple values exceeded the size of the knapsack size m=6.

→ Apply the purging rule on (3,5) and (5,4)

$(3,5) \xi (5,4)$

$(P_j, W_j) \xi (P_k, W_k)$

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$(P_j < P_k)$ and $(w_j > w_k)$

$(3 < 5)$ and $(5 > 4)$ so discard $(3,5)$ tuple.

$S^3 = \{(0,0), (1,2), (2,3), (5,4), (6,6)\}$

Step-3:

Decide which items are placed in the bag

$(P_i w_i) = S^n$ & $(P_i w_i) \notin S^{n-1}$

$n = 3$

$(6,6) \in S^3$ and $(6,6) \notin S^2$

T & T = T

$x_n = 1 = x_3 = 1$

$(P_3, w_3) = (5,4)$

$(6,6) - (5,4) = (1,2)$

$(1,2) \in S^2$ & $(1,2) \notin S^1$

T & F

$x_2 = 0$.

$(1,2) \in S^1$ & $(1,2) \notin S^0$

T & T = T

$(x_1, x_2, x_3) = (1,0,1)$

$\sum P_i x_i = P_1 x_1 + P_2 x_2 + P_3 x_3$

$= 1 \cdot 1 + 2 \cdot 0 + 5 \cdot 1$

$= 1 + 0 + 5$

$= 6$.

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* Travelling salesperson problem:

→ It is to find a tour of minimum cost. A tour of graph 'G' is a directed simple cycle that includes every vertex in 'V'.

→ The cost of the tour is the sum of the cost of the edges on the tour.

→ Suppose we have to route a postal van to pick up mail from mail boxes located at 'n' different sites.

→ One vertex represents the post office from which the postal van starts and to which it must return.

→ Edge (i, j) is assigned a cost equal to the distance from site 'i' to site 'j'.

→ The route taken by postal van is a tour and we are interested to find a tour of minimum length.

→ We can solve travelling salesperson problem by using dynamic programming method.

→ It uses the following computational formulae..

$$g(i,s) = \min_{j \in s} \{ c_{ij} + g(j, s-\{j\}) \}$$

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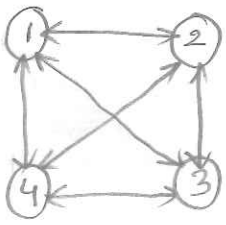
$g(i, \phi) = c_{i1}, \quad 1 \le i \le n$

where $g(i, S)$ be the length of shortest path starting at vertex i going to all vertices in S and terminating at vertex 1.

Problem:

1) consider the directed graph and its adjacent matrix and find tour of minimum cost of travelling sales person problem.

Sol:-



$$\begin{bmatrix} 0 & 10 & 5 & 20 \\ 5 & 0 & 9 & 10 \\ 6 & 13 & 0 & 12 \\ 8 & 8 & 9 & 0 \end{bmatrix}$$

Initially

$g(i, \phi) = c_{i1} \quad 1 \le i \le 4$

$g(1, \phi) = c_{11} = 0$

$g(2, \phi) = c_{21} = 5$

$g(3, \phi) = c_{31} = 6$

$g(4, \phi) = c_{41} = 8$

→ Take $|s| = 1$ that means set S contains only one vertex. From one vertex we can visit all the vertices 2, 3, 4

$$g(i, s) = \min_{j \in S} \{c_{ij} + g(j, s - \{j\})\}$$

$$g(2, \{3\}) = \min_{j \in S} \{c_{23} + g(3, \{3\} - \{3\})\}$$

$$= \min_{j \in S} \{9 + g(3, \phi)\}$$

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$= \min\{9+6\}$

$= 15$

$g(2,\{4\}) = \min_{j\in\{4\}} \{c_{24} + g(4, \{4\}-\{4\})\}$

$= \min\{10+g(4,\phi)\}$

$= \min\{10+8\}$

$= 18$

$g(3,\{2\}) = \min_{j\in\{2\}} \{c_{32} + g(2,\{2\}-\{2\})\}$

$= \min_{j\in\{2\}} (13+5)$

$= 18$

$g(3,\{4\}) = \min_{j\in\{4\}} \{c_{34} + g(4,\{4\}-\{4\})\}$

$= \min\{12+g(4,\phi)\}$

$= 12+8$

$= 20.$

$g(4,\{2\}) = \min_{j\in\{2\}} \{c_{42} + g(2,\{2\}-\{2\})\}$

$= \min\{8+g(2,\phi)\}$

$= 13$

From all the

$g(4,\{3\}) = \min\{c_{43} + g(3,\{3\}-\{3\})\}$

$= \min\{9+g(3,\phi)\}$

$= 15.$

→ Take $|s| = 2$ that means set s contains only two vertices.

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$$g(2,\{3,4\}) = \min_{j\in\{3,4\}} \{c_{23}+g(3,\{3,4\}-\{3\}),\ c_{24}+g(4,\{3,4\}-\{4\})\}$$

$$= \min\{9+g(3,\{4\}),\ 10+g(4,\{3\})\}$$

$$= \min\{9+20,\ 10+15\}$$

$$= 25.$$

$$g(3,\{2,4\}) = \min_{j\in\{2,4\}} \{c_{32}+g(2,\{2,4\}-\{2\}),\ c_{34}+g(4,\{2,4\}-\{4\})\}$$

$$= \min\{13+g(2,\{4\}),\ 12+g(4,\{2\})\}$$

$$= \min\{13+18,\ 12+13\}$$

$$= 25$$

$$g(4,\{2,3\}) = \min_{j\in\{2,3\}} \{c_{42}+g(2,\{2,3\}-\{2\}),\ c_{43}+g(3,\{2,3\}-\{3\})\}$$

$$= \min\{8+g(2,\{3\}),\ 9+g(3,\{2\})\}$$

$$= \min\{8+15,\ 9+18\}$$

$$= 23$$

$$g(1,\{2,3,4\}) = \min_{j\in\{2,3,4\}} \{c_{12}+g(2,\{2,3,4\}-\{2\}),\ c_{13}+g(3,\{2,3,4\}-\{3\}),$$
$$c_{14}+g(4,\{2,3,4\}-\{4\})\}$$

$$= \min\{10+g(2,\{3,4\}),\ 15+g(3,\{2,4\}),\ 20+g(4,\{2,3\})\}$$

$$= \min\{10+25,\ 15+25,\ 20+23\}$$

$$= \min\{35, 40, 43\}$$

$$= 35$$

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path of the four with minimum cost. 35

$c_{12} + g(2, \{3,4\})$

$= c_{12} + c_{24} + g(4, \{3\})$

$= c_{12} + c_{24} + c_{43} + g(3, \phi)$

$= c_{12} + c_{24} + c_{43} + c_{31}$

$= 10 + 10 + 9 + 6$

$= 35$

path is $1 \to 2 \to 4 \to 3 \to 1$

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